

RUHR-UNIVERSITÄT BOCHUM

GLOBAL CHANGE HYDROLOGY (GCH)

Water Quality Scenarios

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What are Scenarios?

It is unlikely that most trends would continue unabated for decades without changing course. [Credit: Munyaradzi Chenje]

- ***„A scenario is a description of a possible event, a sequence of events, or a course of action that could happen in the future.“***
- ***A scenario explores multiple alternative futures based on different driving forces to test how resilient a system is.***

A scenario is not a prediction!

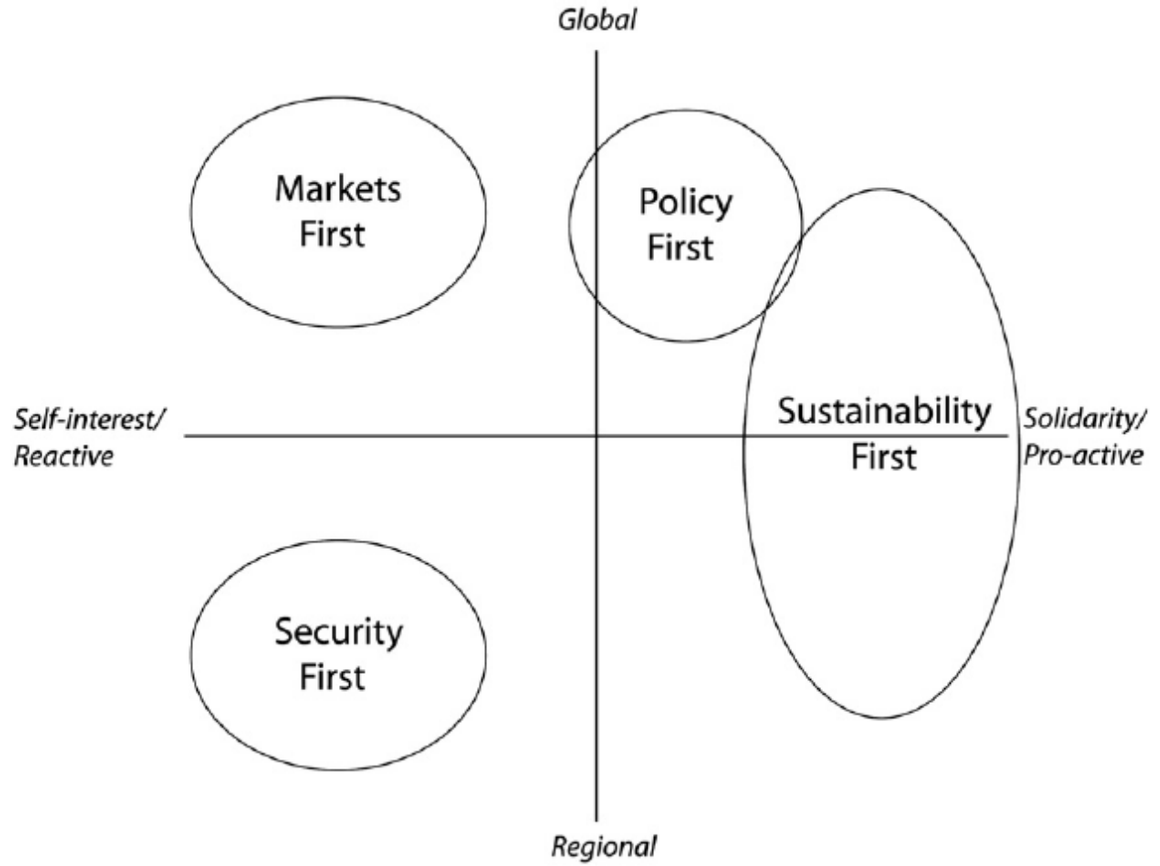
A prediction asks what the most likely outcome is and tries to guess a single future event or number.

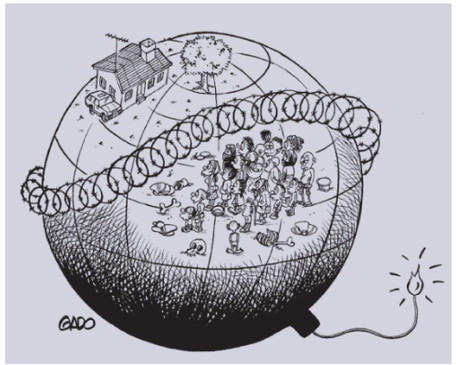
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Exploring the Future





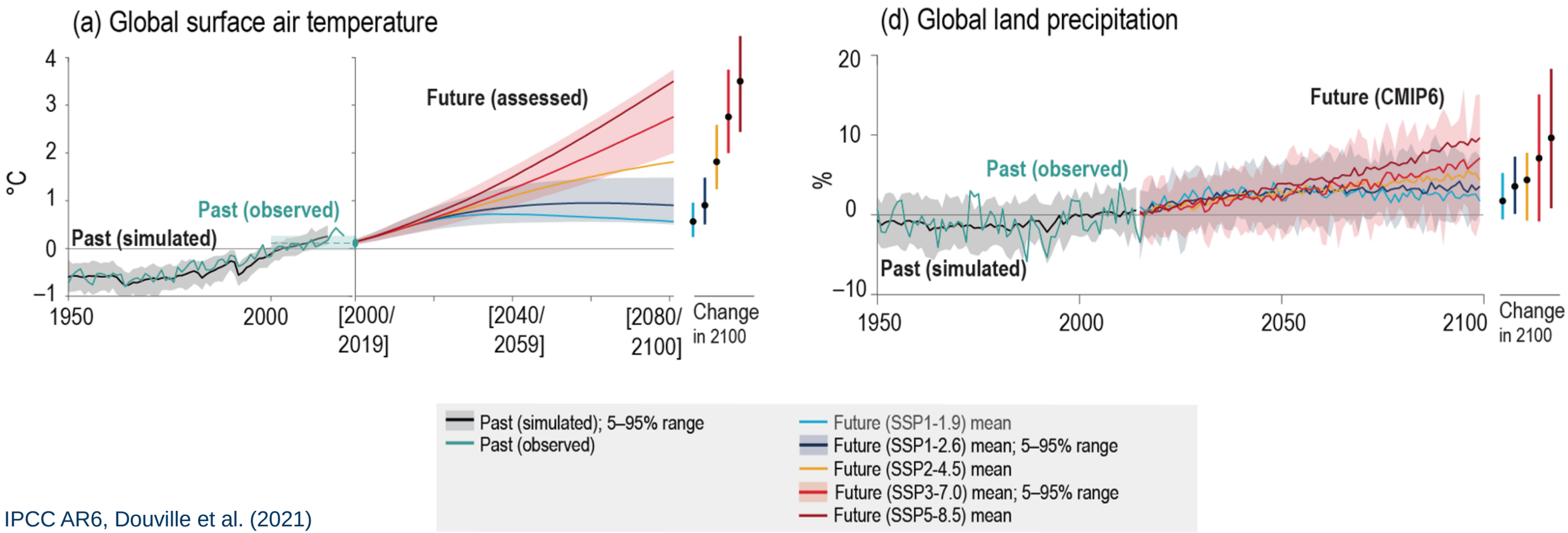






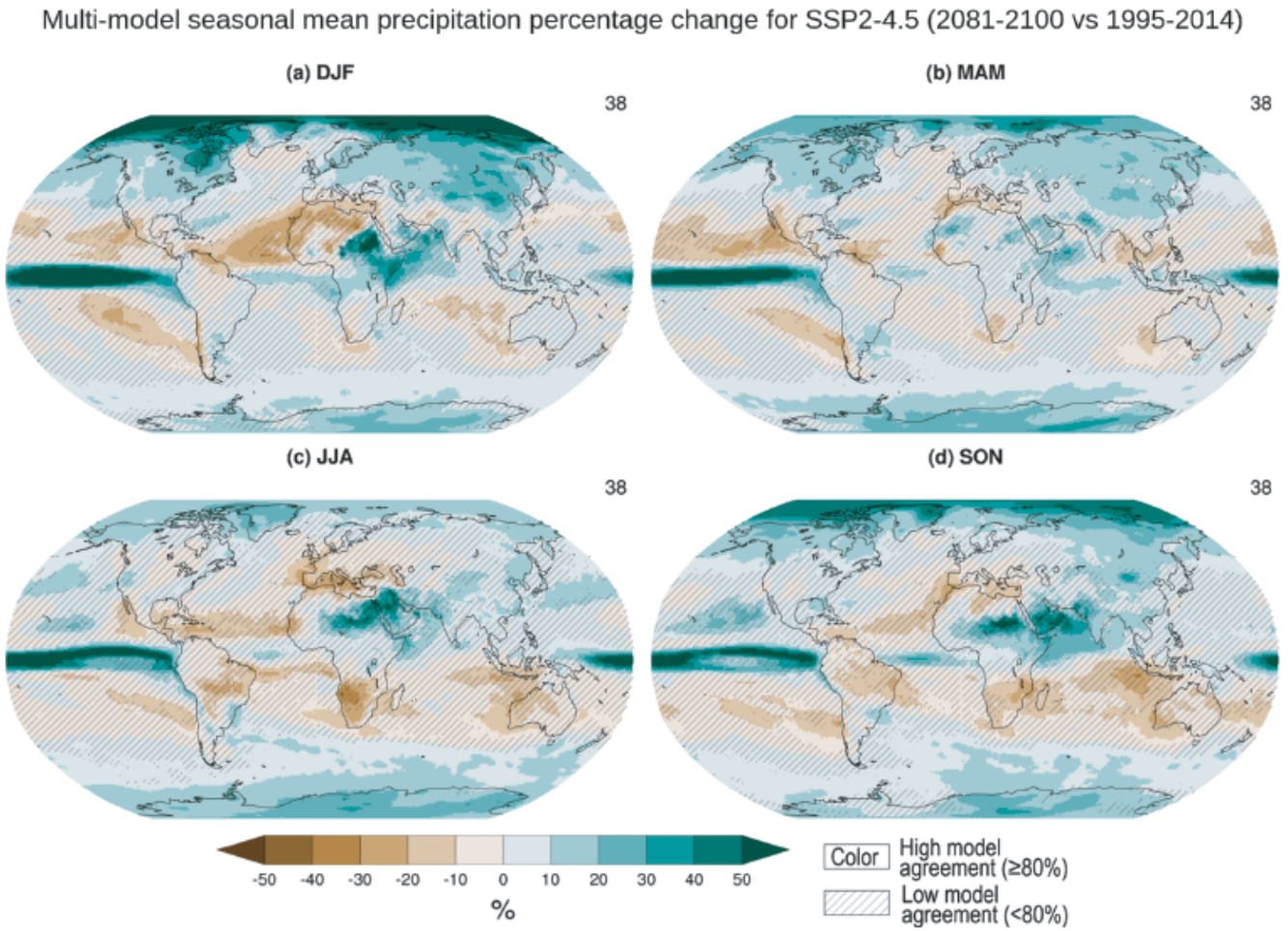
UNEP (2007); Kok et al. (2011)

Climate Change Scenarios



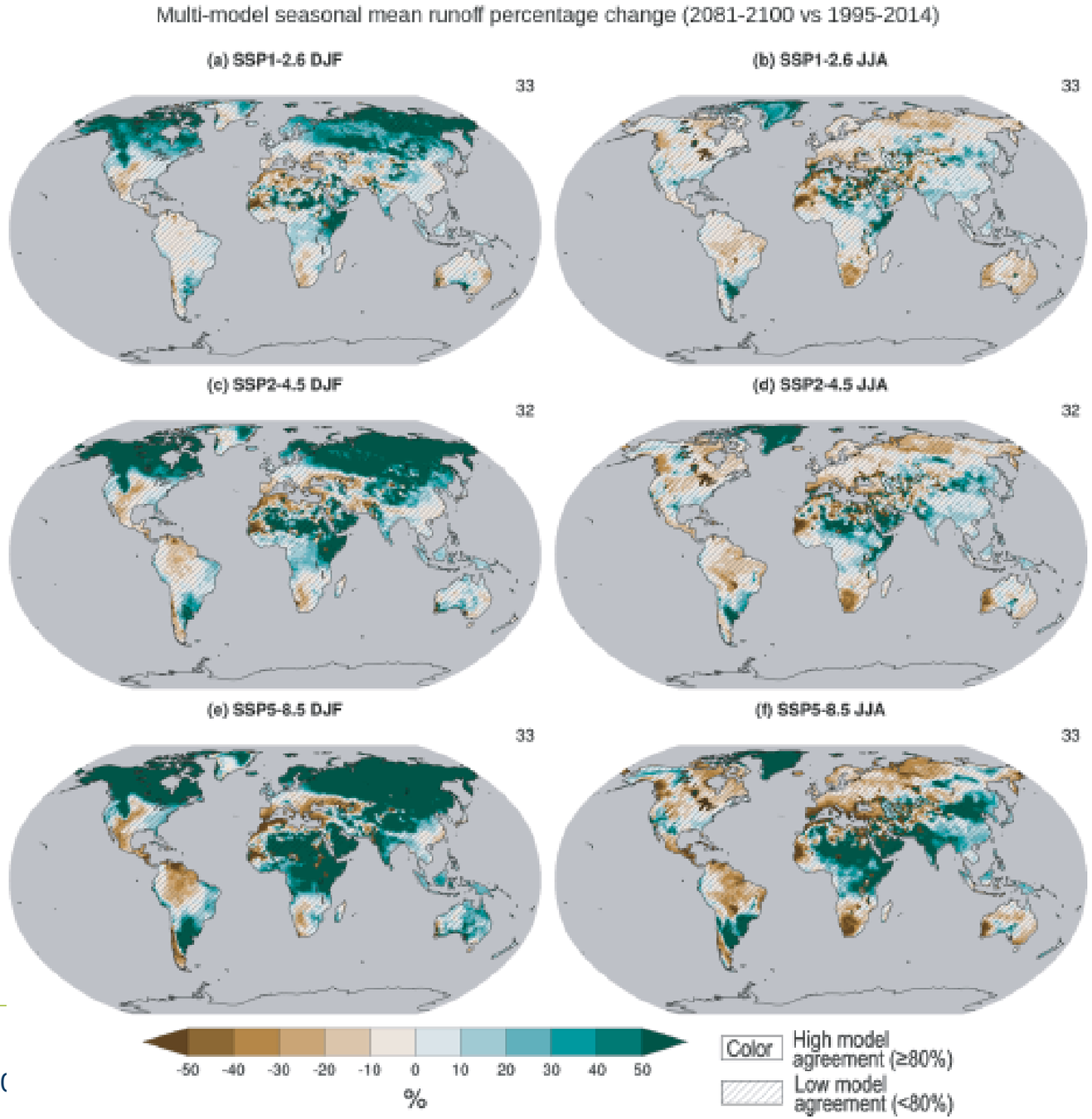
IPCC AR6, Douville et al. (2021)

Future long-term relative changes in seasonal mean precipitation



IPCC AR6, Douville et al. (2021)

Future long-term relative change in mean runoff



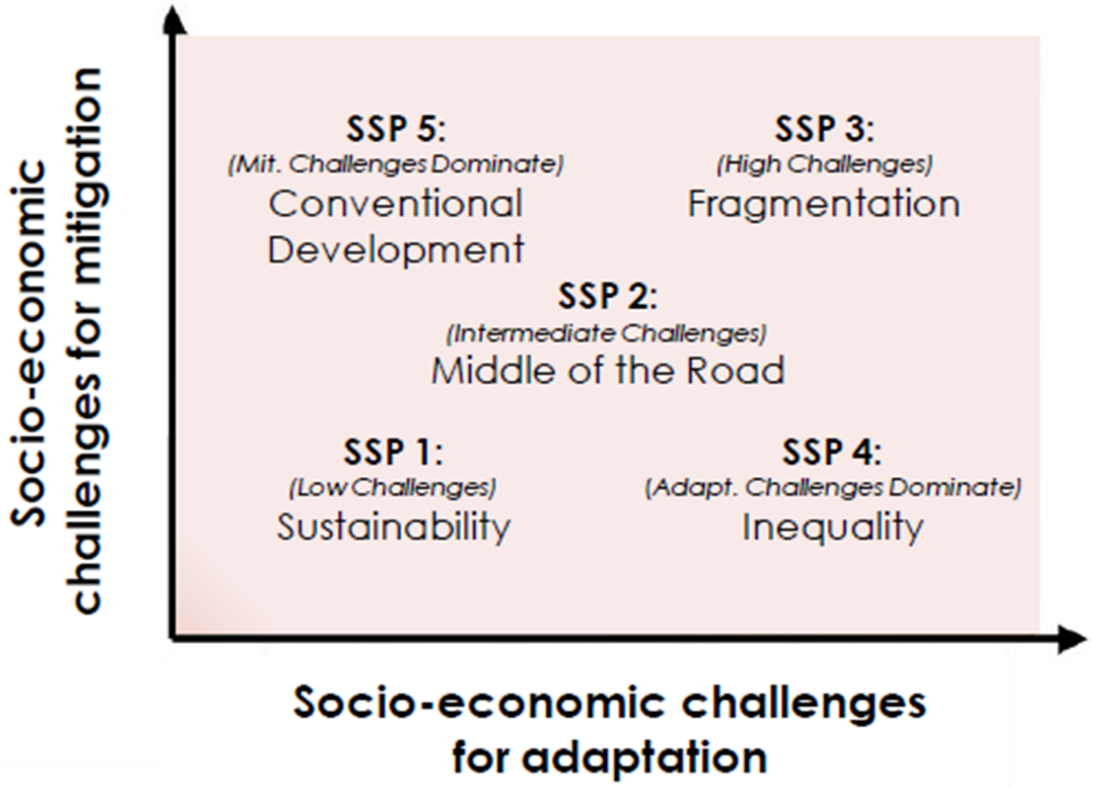
IPCC AR6, Douville et al. (2021)

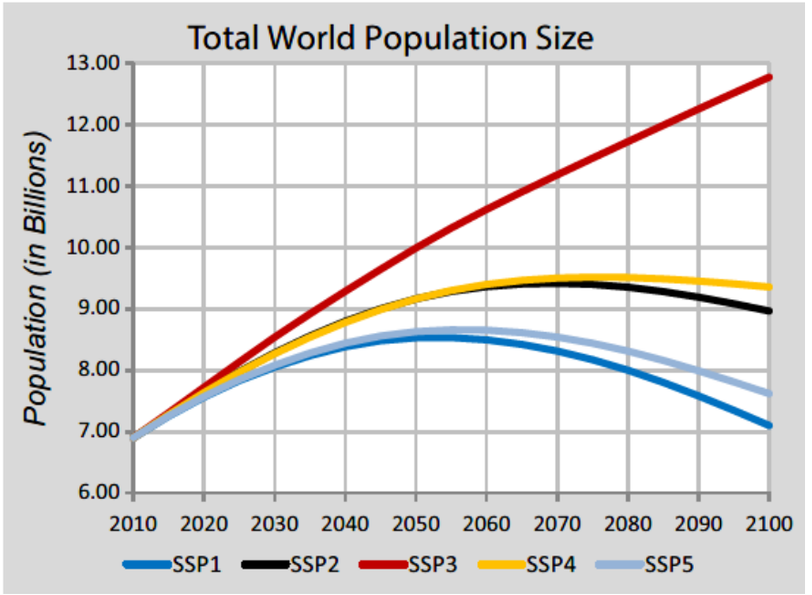
Motivation for Water Quality Scenarios

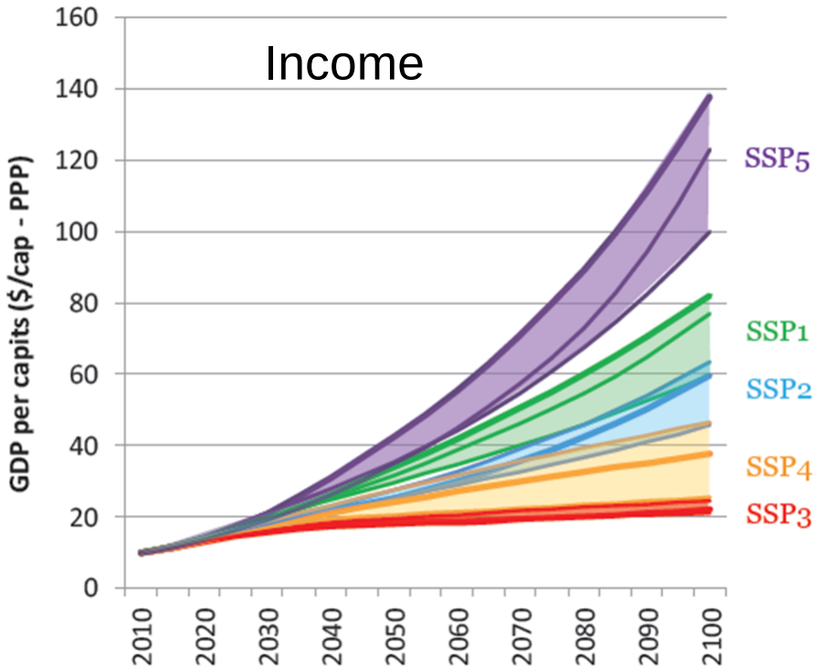
- **SDG6.3** *“By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally”*.
- **Towards a Worldwide Assessment of Freshwater Quality (UN-Water 2016)** *“A scenario analysis that examines dynamic trends over the next decades in water quality and inland fisheries due to various socioeconomic driving forces.*
- **Water Quality Baseline (WWQA 2021)** *“First global display of a water quality and its impacts on ecosystem health, human health and food security as a result from networking activities.”*
- **IPCC (2022)** *“Water quality is affected by climate change. Water quality is briefly mentioned and the report refers to several local studies for a small number of water quality variables.”*

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Future Socio-Economic Development



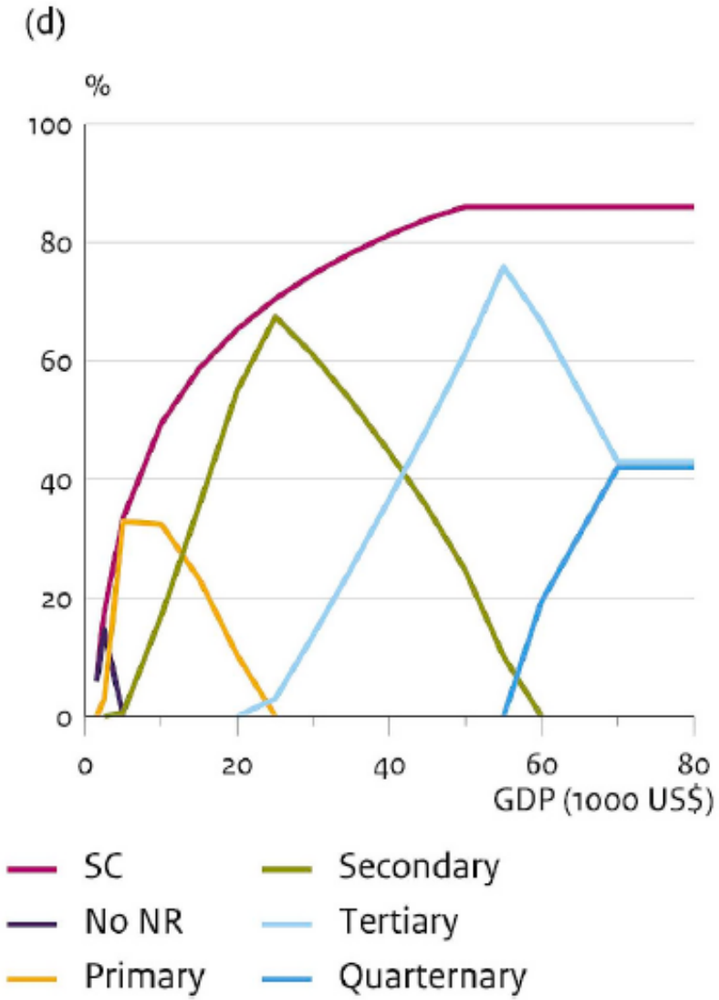


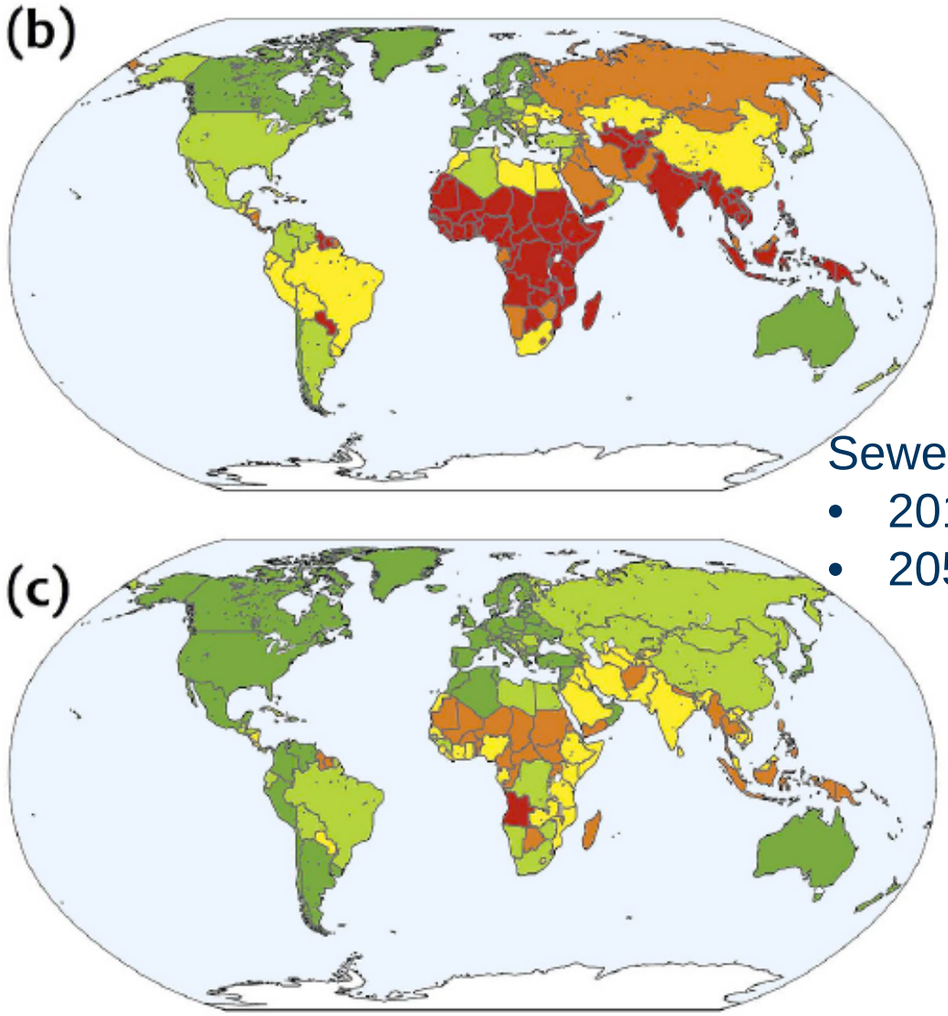


O'Neill et al. (2014)
van Vuuren et al. (2015)
KC & Lutz (2017)
Dellink et al. (2017)
O'Neill et al. (2017)

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Sanitation projections





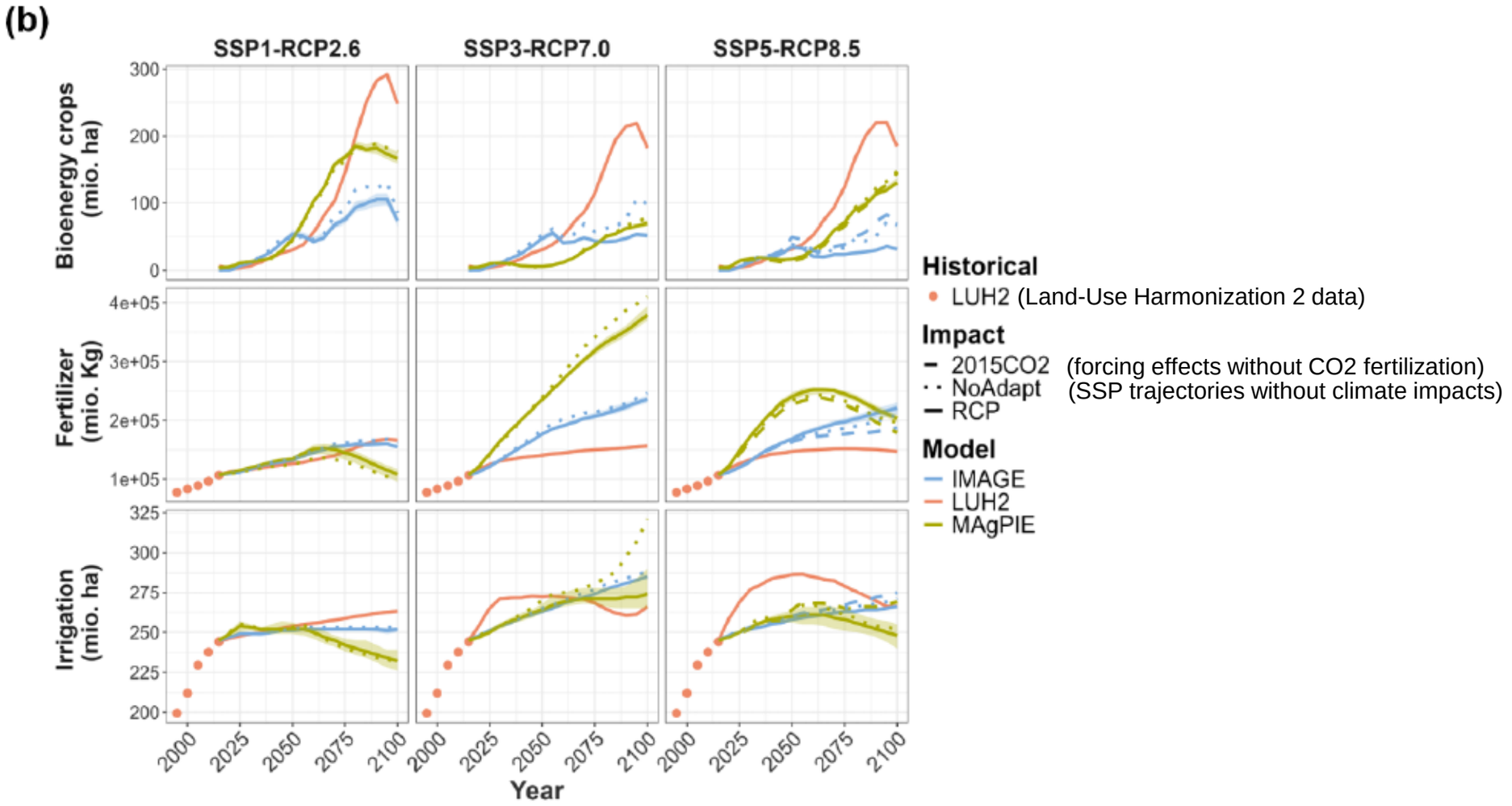
Sewerage connection per country

- 2010 (b)
- 2050 (c)

Van Puijenbroek et al. (2019)

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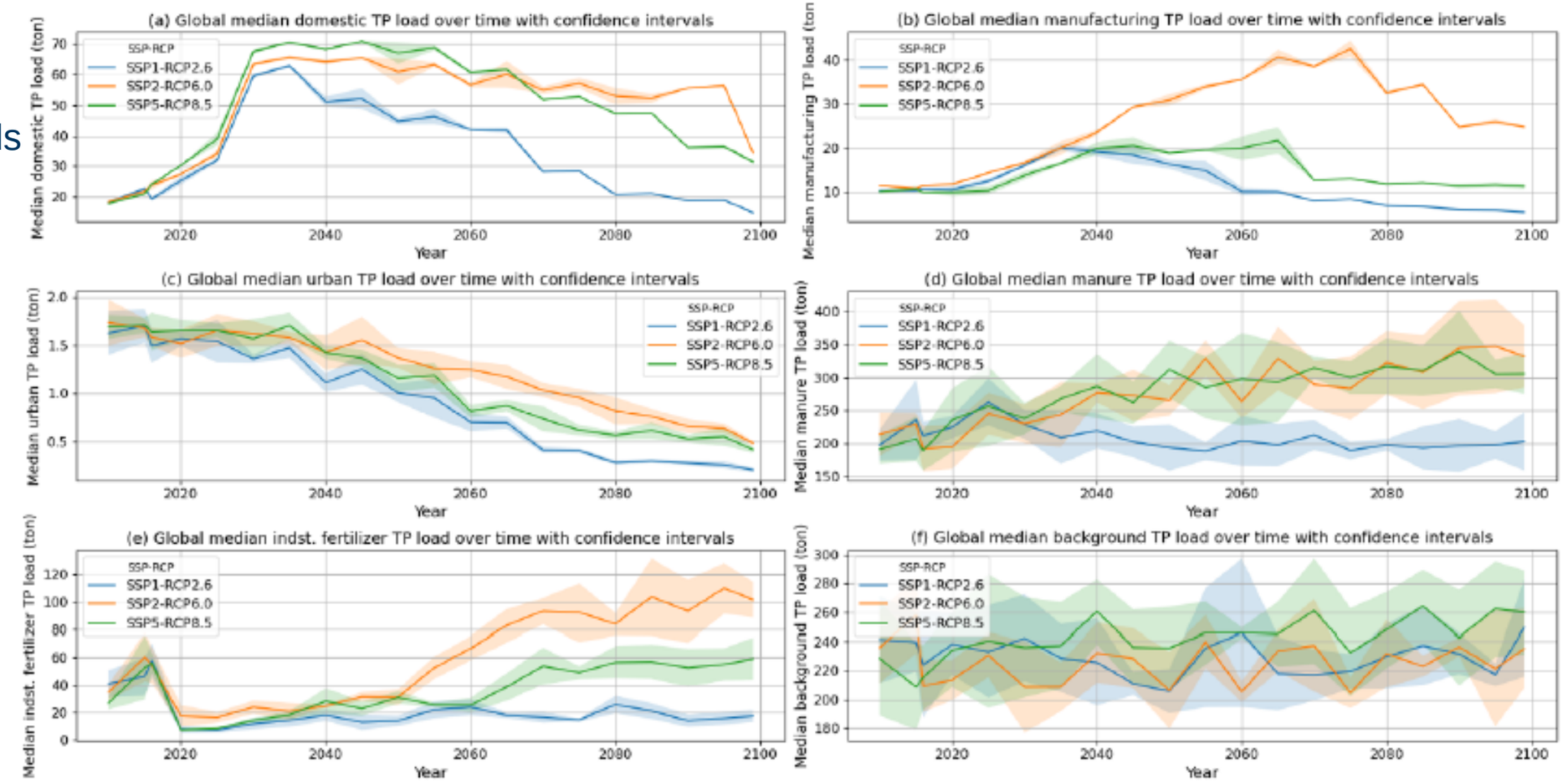
Future Land Use



Global median TP load development

- Manure and background loads consistently emerge as the largest contributors in all scenarios
- urban surface runoff sector remains the smallest contributor

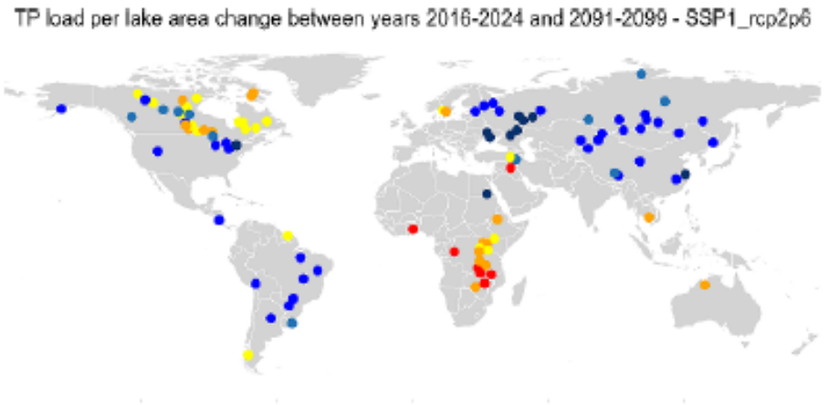
- SSP1-RCP2.6 consistently shows the lowest TP load values throughout the 21st century

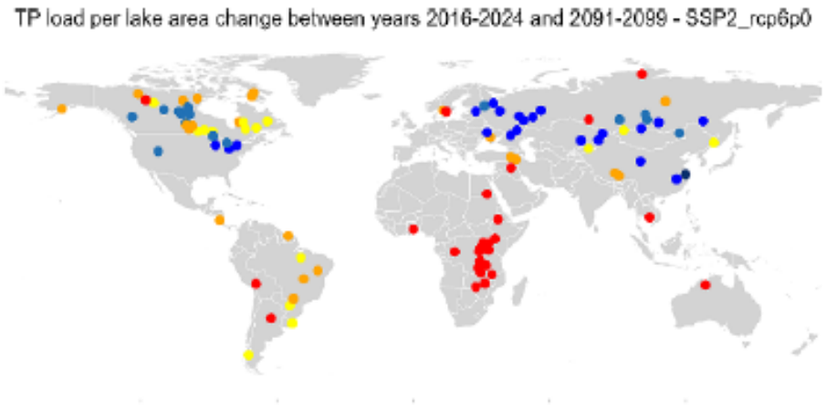


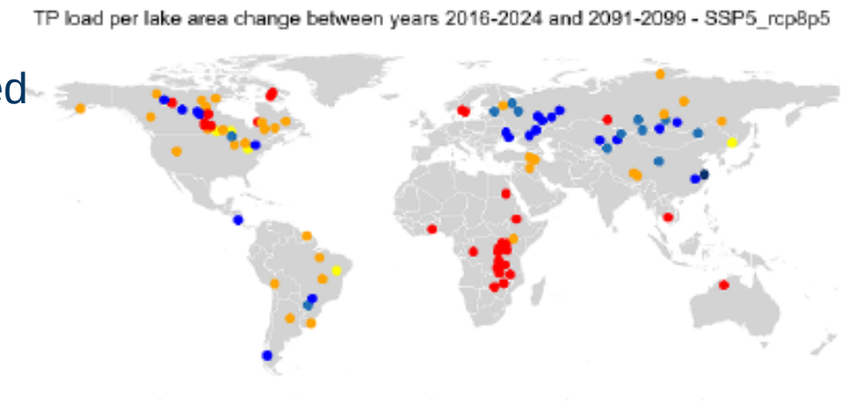
Tilahun et al. (2025)

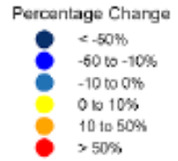
Relative change in TP load per lake area

- North America, Europe, and parts of Asia generally experience a reduction in TP loads
- African lakes, except for Lake Nasser, will be subject to an increase in TP load per lake area
- South American lakes are expected to have a decline in TP load (except Buenos Aires and Blommestein Lakes)



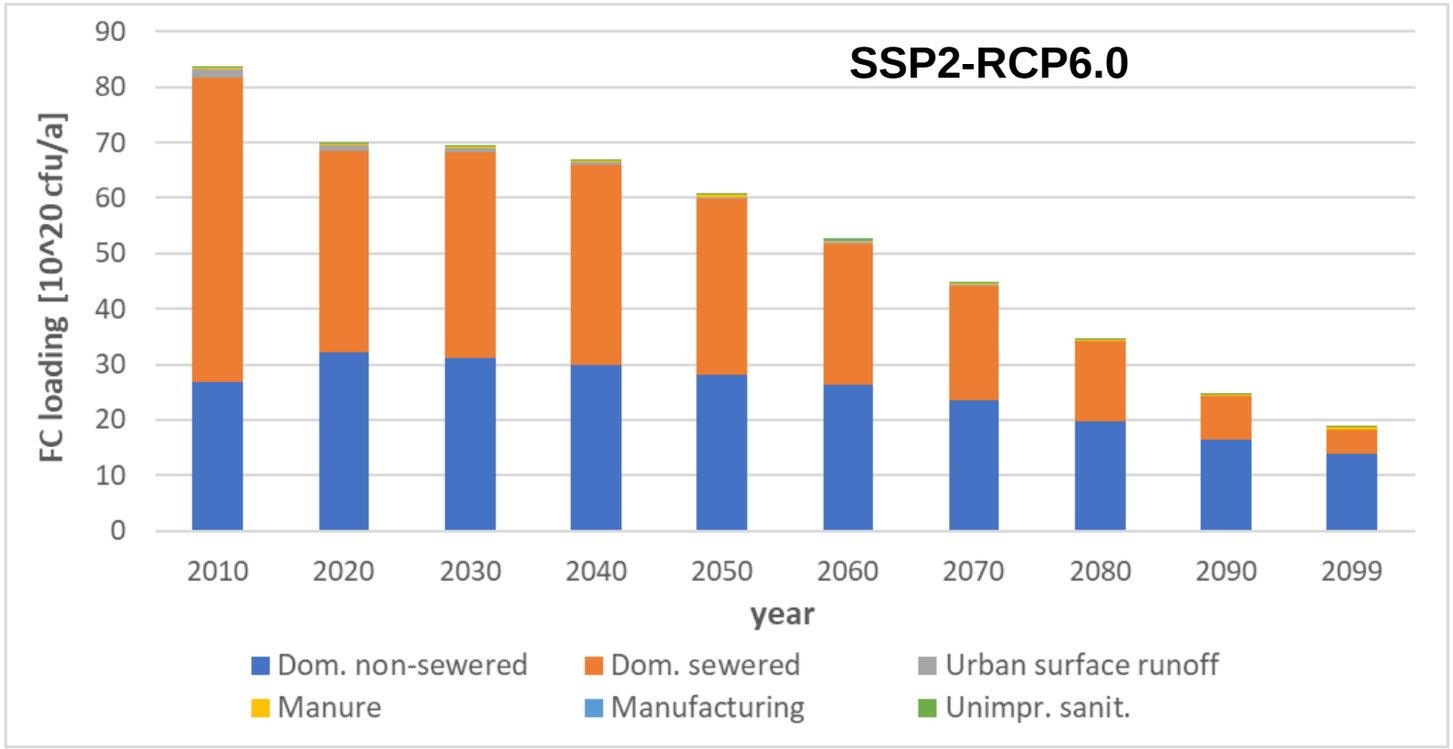




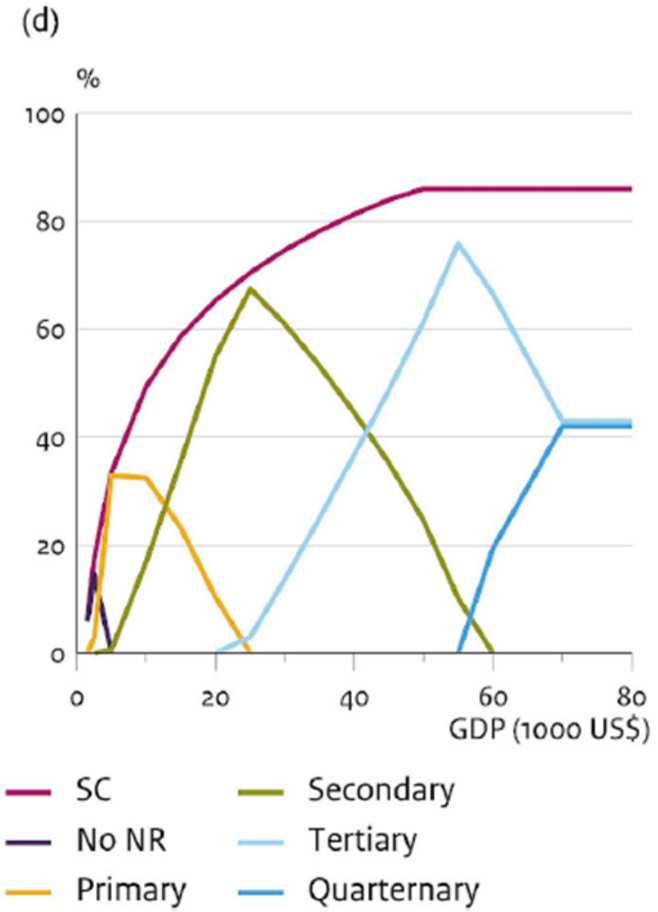


Tilahun et al. (2025)

Future FC loadings



GlobeWQ (2023)

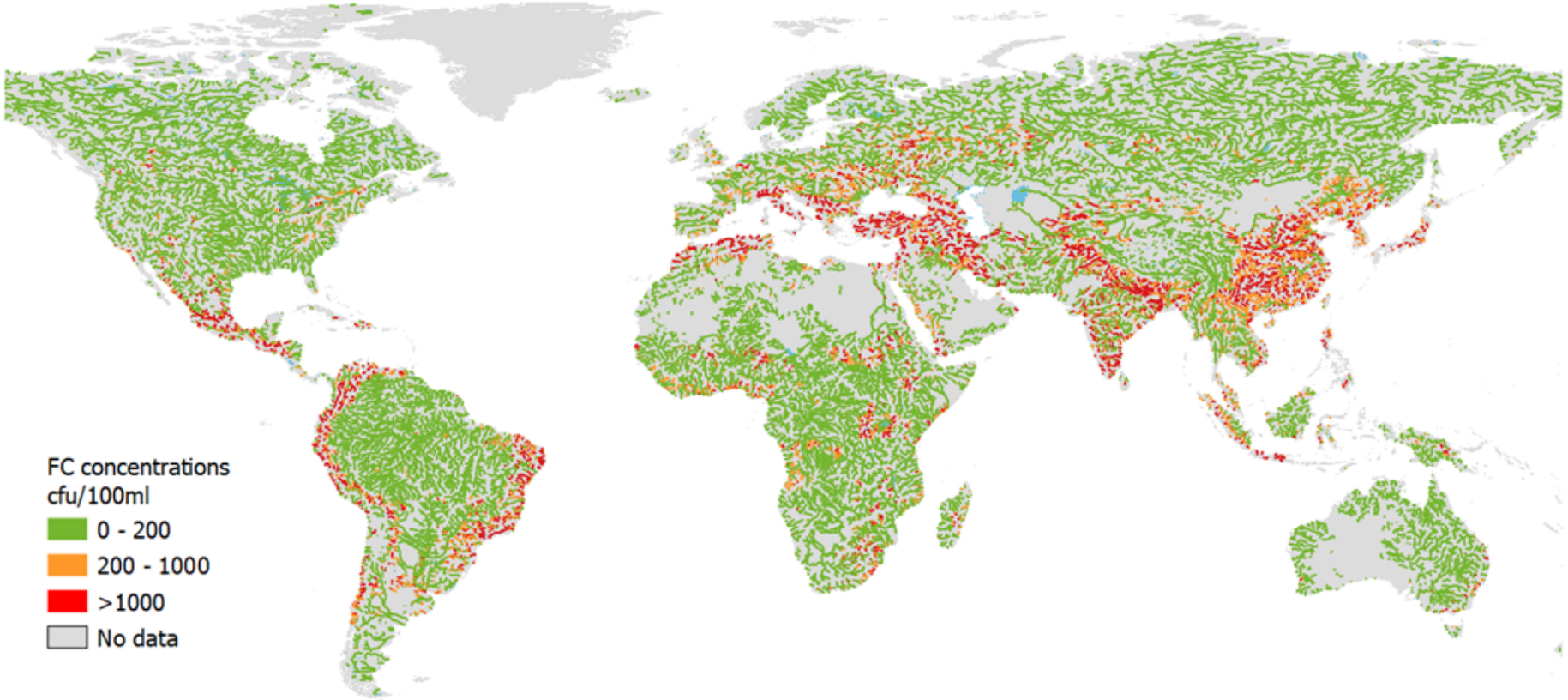


van Puijenbroek et al. (2019)

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FC concentrations in rivers in 2030 (median)

SSP2-rcp6.0, GCM models: GFDL-ESM2M,HadGEM2-ES,IPSL-CM5A-LR,MIROC5

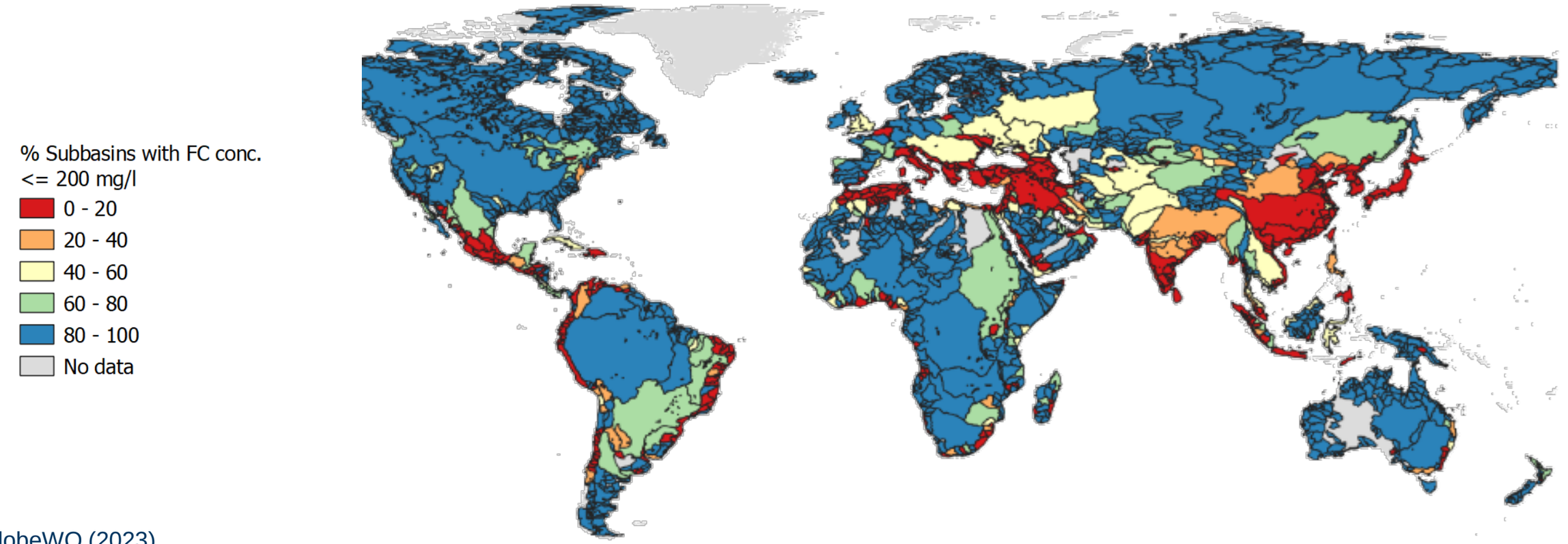


GlobeWQ (2023)

- Improved sanitation reduces FC concentrations in Asia, USA, Europe
- Sanitation improves in almost all African countries beyond 2030
- Pace of sanitation improvement is too slow to become visible in Latin America

Indicator SDG 6.3.2 – FC in 2030

Indicator 6.3.2: “Proportion of bodies of water with good ambient water quality”



GlobeWQ (2023)